

WTW 158 Worksheet 3 on functions, inverses, exponentials and logarithms

A. STUDY GUIDE PROBLEMS

p 33 no 2, 3, 4 p42 no 3, 17, 23, 35 p53 no 17, 19
p66 no 21, 31 p 66 no 23, 38, 41, 51, 56

B. MORE PROBLEMS

1. Sketch the function and determine the domain and range of the function.

a. $m(x) = \ln(1+x)$ b. $n(x) = 1 - \ln x$ c. $h(x) = \ln|x|$ d. $l(x) = |\ln x|$

2. Sketch the function and determine the domain and range of the function.

a. $m(x) = e^{1+x}$ b. $n(x) = 1 - e^{-x}$ c. $h(x) = e^{|x|}$ d. $l(x) = |e^x|$

3. Solve for x :

(i) $\ln(2x^2 + x) \leq 0$ (ii) $|\ln x| \leq 1$

(iii) $\ln|x| \leq 0$ (Use a graph) (iv) $e^x(x-1) > 0$

(v) $e^{x-1} \ln x > 0$ (vi) $e^x \ln|x| > 0$

4. If $f(x) = \frac{1}{x-1}$ and $g(x) = \ln x$, find $f \circ g(x)$ and determine the domain of $f \circ g$.

5. For $f(x) = e^{1+x}$ (2a. above), find the inverse function f^{-1} and sketch both f and f^{-1} .

C. TEST QUESTIONS

Question 1 : If $2 \ln x^3 - 4 \ln x + \ln 5x = \ln a$, then $a =$

[a] $2x^3 + x$	[b] $5x^3$	[c] $\frac{x}{5}$	[d] None of these
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Question 2 : The solution set of $\ln(x^2) - 6 > 0$ is

[a] (e^3, ∞)	[b] $(-\infty, e^3)$	[c] $(-\infty, -e^3)$	[d] $(-\infty, -e^3) \cup (e^3, \infty)$
[e] $(-e^3, e^3)$	[f] None of these		

Question 3: If $xe^{-x} + 2e^{-x} = 0$, then

(a) $x = -2$	(b) $x = -2$ or / of $x = 0$	(c) $x = 0$ or / of $x = 1$
(d) $x = 0$	(e) $x = -2$ or / of $x = 1$	(f) None of these

Question 4: The domain of the function $f(x) = \frac{1}{1 + \ln(x+1)}$ is

(a) $\left(-1 + \frac{1}{e}, \infty\right)$	(b) $(-1, \infty)$	(c) $\left(-1, -1 + \frac{1}{e}\right) \cup \left(-1 + \frac{1}{e}, \infty\right)$
(d) $\left(-1, -1 + \frac{1}{e}\right)$	(e) $x \in \mathbb{R}, x \neq -1 + \frac{1}{e}$	(f) None of these

Question 5: Find the domain of $f(x) = \frac{\ln(x+1)}{(x-1)^2}$.

D YOU TUBE videos: Natural base (e) and Natural logarithm (ln) Kahn Academy
Properties of logarithms Part 1 and Part 2 Patrickjmt

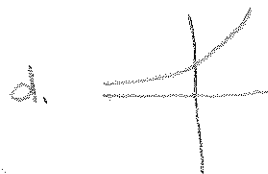
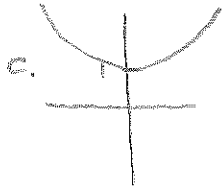
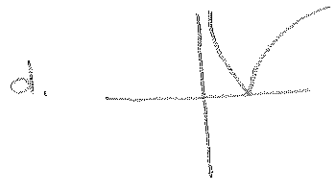
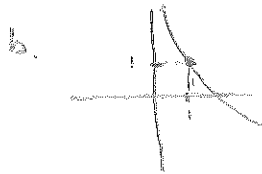
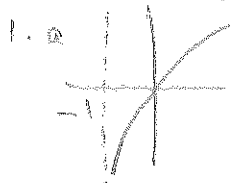
SELECTED ANSWERS:

B. 3 (i) $\left[-1, -\frac{1}{2}\right) \cup \left(0, \frac{1}{2}\right]$ (ii) $\left[\frac{1}{e}, e\right]$ (iii) $[-1, 1] \setminus \{0\}$ (iv) $(1, \infty)$ (v) $(1, \infty)$ (vi) $(-\infty, -1) \cup (1, \infty)$

4. $f \circ g(x) = \frac{1}{\ln x - 1}$ Domain: $(0, \infty) \setminus \{e\}$ 5. $y = \ln x - 1$

C. Q1. [b] Q2. [d] Q3 (a) Q4 (e) Q5 $[-1, \infty) \setminus \{1\}$

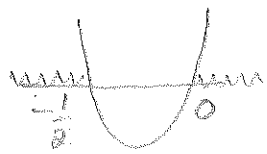
B More problems



3. i) $\ln(2x^2+x) \leq 0$

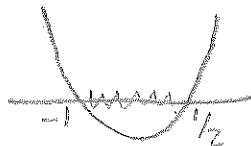
$$\Rightarrow \ln(x(2x+1)) \leq 0$$

$$\Rightarrow x(2x+1) > 0 \text{ and } x(2x+1) \leq 1$$



$$\Rightarrow 2x^2+x-1 \leq 0$$

$$\Rightarrow (2x-1)(x+1) \leq 0$$



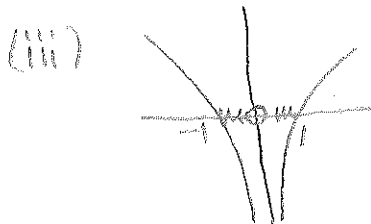
$$\Rightarrow x < -\frac{1}{2} \text{ or } x > 0 \text{ and } x \in [-1, \frac{1}{2}]$$



$$\Rightarrow x \in [-1, \frac{1}{2}] \cup (0, \frac{1}{2}]$$

(ii) $-1 \leq \ln x \leq 1$

$$\Rightarrow e^{-1} \leq x \leq e \Rightarrow x \in [\frac{1}{e}, e]$$



$$x \in [-1, 1] \setminus \{0\}$$

(iv) $e^x(x-1) > 0 \Rightarrow x > 1$

$$\Rightarrow x \in (1, \infty)$$

(v) $e^{x-1} \cdot \ln x > 0$

$$\Rightarrow \ln x > 0 \Rightarrow x > 1$$

$$\Rightarrow x \in (1, \infty)$$

(vi) $e^x \ln|x| > 0$

$$\Rightarrow \ln|x| > 0 - \text{See (iii)}$$

$$\Rightarrow x \in (-\infty, -1) \cup (1, \infty)$$

4. $f(x) = \frac{1}{x-1}$, $g(x) = \ln x$

$$\Rightarrow f \circ g(x) = \frac{1}{\ln x - 1}$$

Domain $x > 0$ and $\ln x - 1 \neq 0$
 $\Rightarrow \ln x \neq 1$
 $\Rightarrow x \neq e$

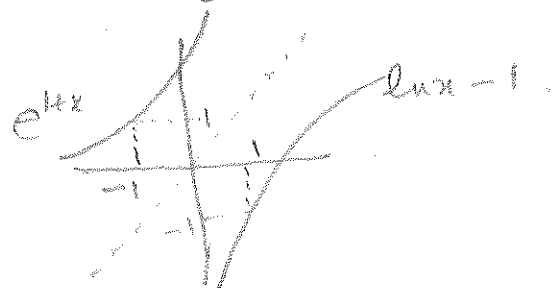
$$\Rightarrow x \in (0, \infty) \setminus \{e\}$$

5. $y = e^{1+x}$

Inverse: $x = e^{1+y}$

$$\Rightarrow 1+y = \ln x$$

$$\Rightarrow y = \ln x - 1$$



C. Test questions

Q1 [b]

Q2 [d]

Q3 [a]

Q4 [c]

Q5 $[-1, \infty) \setminus \{1\}$

A. p 33

2(a) Exponential (b) Power (c) Polynomial

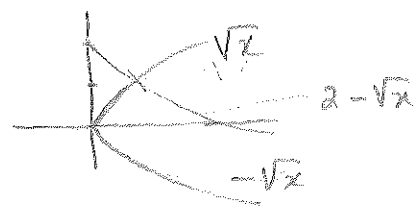
(d) Trigonometric (e) Rational (f) Algebraic

3.(a) - h (b) - f (c) - g

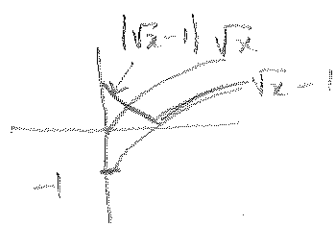
4.(a) - g (b) - f (c) - F (d) - g

42 3(a) - ⑤ (b) - ① (c) - ④ (d) - ⑤ (e) - ②

17. $y = 2 - \sqrt{x}$



23. $y = |\sqrt{x} - 1|$



35. $f(x) = \sqrt{x+1}$, $g(x) = 4x-3$

$f \circ g(x) = \sqrt{4x-2}$; $D_{f \circ g} = [\frac{1}{2}, \infty)$

$g \circ f(x) = 4\sqrt{x+1}-3$, $D_{g \circ f} = [-1, \infty)$

$f \circ f(x) = \sqrt{\sqrt{x+1}+1}$ $x \geq -1$ and $\sqrt{x+1}+1 \geq 0$
 $\Rightarrow \sqrt{x+1} \geq -1$
 $\Rightarrow D_{f \circ f} = [-1, \infty)$

$g \circ g(x) = 4(4x-3)-3 = 16x-15$

p 53

17. (a) $y = e^x - 2$ (b) $y = e^{x-2}$ (c) $y = -e^x$
 (d) $y = e^{-x}$ (e) $y = -e^{-x}$

19. (a) $1 - e^{1-x^2} \neq 0$
 $\Rightarrow e^{1-x^2} \neq 1$
 $\Rightarrow 1-x^2 \neq 0$
 $\Rightarrow x^2 \neq 1, 1$
 $\Rightarrow x \in \mathbb{R} \setminus \{-1, 1\}$

(b) $e^{\cos x} \neq 0 \Rightarrow x \in \mathbb{R}$

p 66

21. $y = 1 + \sqrt{2+3x}$

For inverse: $x = 1 + \sqrt{2+3y}$
 $\Rightarrow (x-1)^2 = 2+3y$
 $\Rightarrow y = \frac{1}{3}(x-1)^2 - \frac{2}{3}$

31. $y = \sqrt{1-x^2}$, $0 \leq x \leq 1$

For inverse: $x = \sqrt{1-y^2}$
 $\Rightarrow x^2 = 1-y^2$
 $\Rightarrow y^2 = 1-x^2$
 $\Rightarrow y = \sqrt{1-x^2}$, $0 \leq y \leq 1$
 f and f^{-1} are the same graphs

b) Circle in first quadrant



p 66 23. $y = e^{2x-1}$

For inverse $x = e^{2y-1}$
 $\Rightarrow \ln x = 2y-1$
 $\Rightarrow 2y = \ln x + 1$

38. (a) $e^{-\ln 2} = \frac{1}{e^{\ln 2}} = \frac{1}{2}$

(b) $e^{\ln(\ln e^3)} = e^{\ln 3} = 3$

41. $\ln(x+2) + \ln \sqrt{x} - \ln(x^2+3x+2)$
 $= \frac{(x+2)\sqrt{x}}{x^2+3x+2}$

51(a) $7-4x = \ln b \Rightarrow 4x = 7 - \ln b$
 $\Rightarrow x = \frac{1}{4}(7 - \ln b)$

(b) $3x-10 = e^2 \Rightarrow 3x = 10 + e^2$
 $\Rightarrow x = \frac{1}{3}(10 + e^2)$

56. $1 < e^{3x-1} < 2$

$\Rightarrow \ln 1 < 3x-1 < \ln 2$
 $\Rightarrow 1 < 3x < \ln 2 + 1$
 $\Rightarrow \frac{1}{3} < x < \frac{1}{3}(\ln 2 + 1)$